ModelIt!

Teacher's Guide

Telephone Game — How Signals Travel

Why Messages Get Lost Along the Way

Quick Reference

Grade Level	Grade NE
NGSS Standard	MS-LS1-8, MS-PS4-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS1-8, MS-PS4-2

Gather and synthesize information that sensory receptors respond to stimuli by sending messages to the brain for immediate behavior or storage as memories. Develop and use a model to describe that waves are reflected, absorbed, or transmitted through various materials.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how signal strength, distance, and noise determine message accuracy.
Disciplinary Core Idea	**LS1.D Information Processing** — Sensory receptors send signals to the brain that are affected by the same degradation factors modeled in this lesson.
Crosscutting Concept	**Cause and Effect** — Students identify that three separate causes (signal strength, distance, noise) all produce effects on the same output (accuracy), sometimes in competing directions.

CAST Testing Alignment

MULTIPLE CHOICE:

A scientist is trying to read brain signals to control a robotic arm. The signals are very weak. Which change would MOST improve accuracy? A) Move the sensors farther from the brain B) Increase noise in the room C) Place sensors closer to the brain D) Use thinner wires

CONSTRUCTED RESPONSE:

Using your model, explain why brain-computer interfaces are so challenging to build. Describe at least two factors that work against signal accuracy and what engineers might do about each one.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every thought, movement, and sensation involves electrical signals traveling through your nervous system. Understanding how these signals work — and what degrades them — is key to building technology that connects brains directly to machines.

How Neural Signals Work

Neurons communicate using electrical impulses called action potentials. These signals travel at speeds of 1-120 meters per second along nerve fibers. Each signal is very small — measured in millivolts (thousandths of a volt). When millions of neurons fire together, their combined electrical activity can be detected on the scalp using electrodes.

The Signal Degradation Problem

As signals travel through tissue, they weaken and spread. The skull and skin between the brain and external sensors act as barriers that reduce signal strength. Additionally, the brain produces billions of signals simultaneously, creating enormous 'noise' that makes it hard to isolate the specific signal you want to read.

Brain-Computer Interfaces Today

Current BCIs range from non-invasive (EEG headsets worn on the scalp) to invasive (electrode arrays implanted directly in the brain). Non-invasive BCIs have low signal strength but are safe. Invasive BCIs have high signal strength but require surgery. Engineers are constantly working to get better signals with less invasive methods.

Why This Matters

BCIs have the potential to restore movement to paralyzed patients, allow locked-in patients to communicate, and treat neurological conditions. The fundamental challenge is always the same: getting a clean, accurate signal from the brain to the machine despite distance, noise, and signal weakness.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Signal Strength, Distance Traveled, and Noise Level as external components (conditions we can set) and Message Accuracy as the single internal component (the outcome determined by all three inputs).

E **Establish Relationships**
Students determine that Signal Strength positively affects accuracy, while Distance and Noise both negatively affect it — three forces acting on one output.

V **Visualize & Model**
Students build a model showing three external arrows pointing into one internal component, with positive and negative relationships.

E **Evaluate Outcomes**
Students run best-case, worst-case, and noise-test scenarios to see how different combinations affect accuracy.

R **Revise & Extend**
Students add Shielding or Number of Channels to explore how engineers fight back against signal degradation.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"What if you could control a robot just by THINKING? That is what brain-computer interfaces try to do. But it is incredibly hard. Today we find out why."*

Watch For

Show a brief video clip of a BCI patient moving a robotic arm with their thoughts (many are available on YouTube). Ask: What made that possible?

Slide 2: Learning Objectives

3 min

Say: *"We are going to model the biggest challenges in sending signals from one place to another."*

Watch For

Have students read objectives. Pre-teach 'signal' and 'noise' with everyday examples (phone static, crowded room).

Slide 3: Big Question

3 min

Say: *"Your brain controls your hand perfectly. But what if your brain had to control a robot across the room? What goes wrong?"*

Watch For

Play a quick telephone game with 8-10 students. The message degrades. Ask: Why?

Slide 4: LEVER Framework

2 min

Say: *"We are going to model this problem to figure out what we can do about it."*

Watch For

Briefly review LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What affects whether a message arrives correctly? Three things work against us — and one helps."*

Watch For

Guide sorting: Three externals (signal strength, distance, noise) and one internal (accuracy). Discuss why this lesson has more externals than internals.

Slide 6: Activity 2: Connections

8 min

Say: *"Signal strength HELPS accuracy — that is a positive arrow. But what about distance and noise?"*

Watch For

Help students draw positive arrow from signal strength and negative arrows from distance and noise. Key: three arrows, one output.

Slide 7: Activity 3: Simulation

10 min

Say: *"Best case: strong signal, close, quiet. Worst case: weak signal, far, noisy. Let us see the difference!"*

Watch For

Run three scenarios with predictions. Key moment: the noisy room scenario — can strength overcome noise?

Slide 8: Discoveries

5 min

Say: *"Now you understand WHY building a brain-computer interface is one of the hardest engineering challenges on Earth."*

Watch For

Summarize findings. Connect to real BCI challenges: brain signals are tiny, skull creates distance, other neurons create noise.

Slide 9: STEM Challenge

12 min

Say: *"Let us experience signal degradation firsthand with a real circuit!"*

Watch For

Distribute circuit materials. Groups build LED circuits and test effect of wire length and buzzer noise.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Signal Strength, Distance Traveled, Noise Level
- **INTERNAL:** Message Accuracy

Relationships

- **Signal Strength to Message Accuracy:** POSITIVE (+) — A stronger original signal is harder to degrade. Like shouting vs. whispering — a shout can be heard even in a noisy room.
- **Distance Traveled to Message Accuracy:** NEGATIVE (-) — Signals weaken as they travel farther. Like how a flashlight gets dimmer the farther away you hold it. More distance = more degradation.
- **Noise Level to Message Accuracy:** NEGATIVE (-) — Noise drowns out the signal. Like trying to hear someone talk at a loud concert — the message gets buried under interference.

Key Discoveries

- Three different factors ALL affect whether a message arrives accurately
- Signal strength HELPS accuracy, but distance and noise HURT it
- Even a strong signal can be ruined by too much noise
- This is exactly the challenge scientists face when trying to read brain signals for brain-computer interfaces

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Brain signals are like Wi-Fi — they broadcast everywhere"	Brain signals are NOT like radio waves. They are tiny electrical impulses that travel along specific nerve fibers, like electricity through a wire. They do not broadcast outward — sensors have to be very close to detect them. This is why BCI sensors need to be on or in the head, not across the room.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"A stronger signal always fixes accuracy problems"	While stronger signals help, they cannot overcome all challenges. If noise is extremely high, even a very strong signal gets drowned out. And increasing signal strength in the brain is not really an option — you cannot tell your neurons to 'fire harder.' Engineers have to work with the signals the brain naturally produces.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Computers can easily separate signals from noise"	While computers are getting better at filtering noise, it remains one of the hardest problems in engineering. The brain produces trillions of signals per second, and the one you want might be a tiny fraction of that. Modern BCI software uses machine learning to get better at finding the signal, but it is still far from perfect.